Detection Method for Cascading Failure of Power Systems Based on Epidemic Model

Jian Xu, *Senior Member*, *IEEE*, Zhonghao He, *Student Member*, *IEEE*, Siyang Liao, *Member*, *IEEE*, Yuanzhang Sun, *Senior Member*, *IEEE*, Liangzhong Yao, *Fellow*, *IEEE*, Deping Ke, and Jun Yang, *Member*, *IEEE*

***Abstract*—The early detection of cascading failure plays an important role in the safe and stable operation of the power system with high penetration of renewable energy. This paper proposes a fault propagation dynamic model based on the epidemic model, and further puts forward a method to detect the development of cascading failures. Through the simulation of the IEEE 39-bus and 118-bus systems, this model is proven to be valid and capable of providing practical technical support for the prevention of cascading failures in power systems with high penetration of renewable energy. This paper also provides an analysis method for the choice of different protection and control measures at each stage of cascading failure, which has critical significance and follow-up value.**

***Index Terms*—Cascading failure, complex network, epidemic model, fault propagation, renewable energy, weight method of game theory.**

I. Introduction

In the context of high proportion of renewable energy and electronic equipment in power systems, the two features of renewable generation, weak support and low immunity, gradually predominates the operation of power systems with high penetration of renewable energy [1]. The randomness, intermittency and volatility of wind power generation and photovoltaic power generation have direct influence on the power flow, causing more complexities and security risks in the operation process. Without proper response, unexpected power failures are very likely to proliferate and even threaten the whole system, incurring considerable losses like the "9.28" blackout in South Australia [2] and the "8.9" blackout in Britain [3]. Therefore, the safe and efficient operation of a power grid is one of the most important and urgent problems remaining to be solved. The cascading failure detection method is necessary to monitor the whole power grid, locate the weak links in fault propagation, and develop suppression control strategy.

Currently, the detection methods of cascading failures are mainly divided into three categories. The first type is based on models, represented by ORNL-PSERC-Alaska (OPA) [4], CASCADE [5] and Motter-Lai model [6]. In [7], the improved OPA model is utilized to reveal the critical characteristics of power failures and assess the safety level of a power system. Reference [8], based on the branching process model, establishes the relationship between topology changes and cascading failures in power system and calculates the average propagation velocities under different topology conditions in accordance with branching process parameters. The second type is based on pattern search, represented by Monte Carlo sampling [9] and fault chain [10]. Reference [11] integrates knowledge graph and machine learning algorithm and proposes an intelligent method to trace and predict characteristic events which further realizes online traceability and prediction of fault chain. In [12], based on Monte Carlo sampling, the severity of cascading failures is described by two comparisons; one is between the voltage of each node and the collapse voltage, and the other is between the power flow of each branch and its limit. The third type is based on risk assessment [13]. References [14] and [15] introduce random power flow and risk value theory to calculate the load loss risk, which provide guidance for the intra-day scheduling and thus helps suppress cascading failure development.

The foregoing research prefers to describe the evolution process of power grid over a mid-to-long term. The mechanism and law of cascading failures are explored through qualitative analyses. However, most of them focus on the path prediction of cascading failures and lack the detection of the severity of cascading failure. In fact, for operators what matters most is exactly the risk of cascading failure of power grid in operation conditions. Hitherto, researchers have not taken enough account of the correlation between different fault components and the interaction between adjacent faults in propagation process. There is also a lack of quantitative evaluation of different stages of cascading failures, as well as the exploration of the impact of low inertia on power systems with high penetration of renewable energy.

Therefore, more in-depth research and discussions should be conducted to explore how to effectively realize situational awareness of cascading failure in power grid with the consideration of renewable generation characteristics.

Cascading failure propagation in a complex power network [16]–[18] obeys certain laws, so it is crucial to model the characteristics of its propagation behavior. At present, some scholars have borrowed the epidemic model to simulate the fault propagation process in power systems. Reference [19]

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J. Xu, Z. H. He, S. Y. Liao (corresponding author, email: liaosiyang@whu.edu.cn; ORCID: https://orcid.org/0000-0002-4092-764X), Y. Z. Sun, L. Z. Yao, D. P. Ke and J. Yang are with the School of Electrical Engineering and Automation, Wuhan University, Wuhan 430072, China.

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establishes a dynamic model to predict the fault propagation behavior of power grid with the assistance of a susceptible-infected (SI) model and analysis of subgraph indices and spatiotemporal correlation of power grid. Reference [20] applies the epidemic model to the study of voltage offset in distribution network and establishes the relationship between infection rate, network topology and voltage offset threshold. Reference [21] analyzes the propagation of frequency oscillation in power grid with susceptive-infected-susceptible (SIS) model and proves the validity of epidemic model in frequency oscillation dynamics propagation. On this basis, [22] investigates the coupling effect of information layer and physical layer on the fault scale in a power cyber physical system (CPS) network. However, among existing epidemic models, the division of the state of cascading failures is oversimplified, and few studies quantify the infection rates and cure rates which serve as the reasons for the change of the power system states.

Based on the epidemic model in epidemic transmission dynamics, this paper analyzes the development process of cascading failure and proposes a detection method. The main contributions of this paper are as follows:

1) This paper innovatively proposes a fault propagation dynamic model based on the SIR epidemic model. The density of infected nodes is used to describe the development severity of cascading failures, which effectively simulates the impact of cascading failures on power system components in the propagation process and map the real-time operating states of the power system to cascading failure risk.

2) The situational awareness method is proposed to provide early warning of cascading failures and distinguish the status of each node, where the low inertia, as well as other electrical features of power systems with high penetration of renewable energy can be fully considered. The advantages of the proposed method over the traditional method and the SI epidemic model are discussed.

3) The propagation rules based on the dynamic model of fault propagation are defined by the fault severity index to realize the simulation of fault propagation. The effectiveness of the proposed method is further verified under systems with different renewable energy penetration rates and different sizes. The early implementation of key strategies can significantly reduce the negative impact of cascading failures based on the simulation results.

The remainder of this paper is organized as follows: Section II proposes a fault propagation dynamic model based on the epidemic model. Section III defines the fault severity index by AHP-entropy weight method. Section IV describes the steps of the fault propagation dynamics through which the infection and cure rate of power systems are obtained. Section V validates the effectiveness of the proposed method on the IEEE 39-bus and 118-bus systems. Finally, conclusions are drawn in Section VI.

II. Fault Propagation Dynamic Model Based on Epidemic Model

Cascading failure refers to a series of failure events where one or more initial faults start successive failures of other components in power systems [23]. After the initial trigger stage, the factors that aggravate the cascading process in the system mainly come from the system itself, including line overload outage caused by power flow transfer, protection malfunction/rejection, hidden failures, and system de-alignment.

In order to prevent cascading failures, the transient stability of power system should be given enough attention. According to the load constraints and operation constraints of the power system, the operation states of the power system can be generally divided into four types: normal state, alert state, emergency state and recovery state [24]. When the power system is in normal operation state, the load is matched with the generation at any time. Otherwise, the fault part of the system should be removed precisely and timely through the cooperation of automatic control, protection devices and operation personnel. If the fault propagation can be perceived in advance and mitigation strategies can be adopted in the critical link, the risk of cascading failures in the power system can be effectively reduced.

For the power system, its physical mechanism is manifested as data. As for other fields, although the physical systems are different, their external manifestations are also expressed as data. When a system is manifested as data, one feasible choice is to refer to the theoretical guidance and analysis methods based on data-driven in other fields in the research of power system.

When a certain fault disturbs a power system, it will first affect a weak node and then spread outwards to other weak nodes. After the disturbance, the protection strategy and the timely treatment by dispatcher will transform those nodes into recovered nodes. The whole process of fault propagation in power system has great similarity to the epidemic propagation process in the medical community [25]. For epidemic propagation, its time behavior needs to be predicted, while for fault propagation, the dynamic behavior should also be predicted. Therefore, the epidemic model can be employed to study the fault propagation process in a power system. The analogy between them is shown in Fig. 1.

A. Epidemic Model

The epidemic model is an important foundation of the power fault propagation model. Researchers have proposed a variety of network transmission models based on the epidemic model, where SI [26], SIS [27] and the susceptible-infected-recovered (SIR) model [28] are widely used. In SIR model, the susceptible node S is transformed to an infected state under the influence of infected node I, while I is cured by itself or external measures and becomes recovered node R. The ratio of infection rate and cure rate is recorded as a basic infection number indicator. The infection rate means that if a susceptible node is adjacent to one or more infected nodes, it will become an infected node at a preset probability, while the cure rate refers to another preset ratio that an infected node becomes an immune one. These rules are executed simultaneously throughout the whole network. The propagation will become more widespread when the infection rate is higher, and the cure rate is lower.

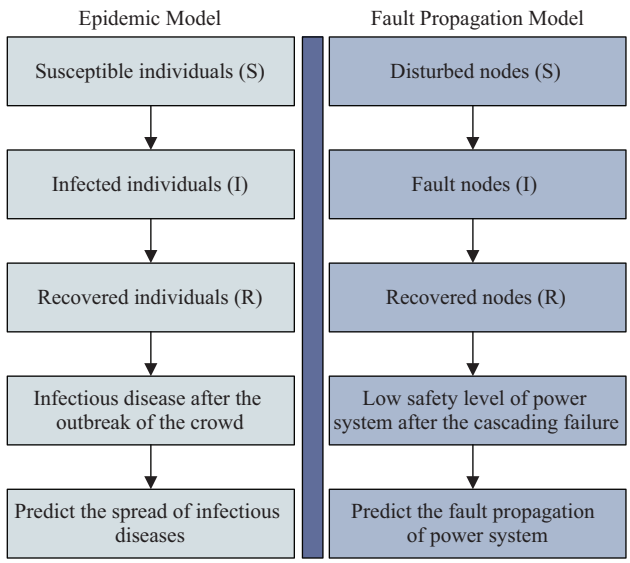

Fig. 1. Comparison of epidemic and fault transmission.

Based on the SIR epidemic model, this paper studies the transmission characteristics of cascading failures. In SIR epidemic model, individuals are divided into three states: S, I and R.

S(Susceptible) – susceptible state. These individuals are healthy but may be infected by viruses.

I(Infected) – infected state. These individuals have been infected by viruses and capable of infecting other healthy individuals.

R(Recovered) – immune state. These individuals are healthy and will not be infected by the virus again.

The assumptions of this are presented below:

1) The total number of individuals in the system remains stable. The influence of the number of newborn, dead and migrated individuals on the total number is trivial, so the total number can be assumed to be constant.

2) If an individual is infected, the individual changes from the susceptible state to the infected state.

3) Through external treatment, infected individuals can change to an immune state, which cannot be infected again.

Assume $s(t)$, $r(t)$ and $i(t)$ respectively represent the individual density of the susceptible, the infected and the immune state at time t. λ represents the probability that S individuals will be infected into I; μ is the probability that I individuals will be cured into R; N is the total number of individuals in the system. Each infected individual will infect $\lambda s(t)$ individuals, and the number of infected individuals is $Ni(t)$, so the number of newly infected individuals is $\lambda N s(t) i(t)$.

Among $Ni(t)$ infected individuals, $\mu N i(t)$ individuals are cured and become recovered individuals. The states of individuals in the system only include S, I and R. Therefore, the dynamic behavior of disease transmission in the SIR epidemic model can be described as the following differential equations:

$$\begin{cases} \frac{\mathrm{d}s(t)}{\mathrm{d}t} = -\lambda s(t) i(t) \\ \frac{\mathrm{d}i(t)}{\mathrm{d}t} = \lambda s(t) i(t) - \mu i(t) \\ \frac{\mathrm{d}r(t)}{\mathrm{d}t} = \mu i(t) \end{cases} \tag{1}$$

B. Dynamic Model of Fault Propagation

The dynamic model of fault propagation describes the state of each node in the current time t. For each node, its own state will change with time according to the established transformation rules and interact with the states of the surrounding nodes.

When establishing the dynamic model of fault propagation, the nodes in power system are divided into three states, S, I and R with reference to the SIR epidemic model.

The relevant parameters of epidemic model and of fault propagation dynamics model are in Table I.

In the fault transmission of power system, if a node is disturbed, it becomes a fault node; if a fault node is cured, it becomes an immune node. The total number of nodes in system is constant. Therefore, the dynamic behavior of fault propagation can be described as the differential equations in (1).

There are three states for all nodes in the system, so

$$i(t) + s(t) + r(t) = 1 \tag{2}$$

Assume that the initial density of susceptible, infected and recovered nodes are $s(0) = s_0$, $i(0) = i_0$, $r(0) = r_0$, and the equations (1) can be solved by phase trajectories [29].

TABLE I
COMPARISON OF PARAMETERS IN TOW MODELS ABOVE

Parameters	Epidemic Model	Fault Propagation Model
S	Susceptible state	Vulnerable nodes
I	Infected state	Fault nodes
R	Recovered state	Recovered nodes
N	The total population	The total nodes
$s(t)$	The individual density of S state at time t	The density of vulnerable nodes at time t
$i(t)$	The individual density of I state at time t	The density of fault nodes at time t
$r(t)$	The individual density of R state at time t	The density of recovered nodes at time t
λ	The infection rate of disease	The infection rate of fault propagation
μ	The cure rate of disease	The cure rate of fault propagation

The basic infected index R_0 is the ratio of infection rate and cure rate. In the epidemic model, this index is an important mark about the popularity of disease, which can be used to distinguish whether the disease is prevalent.

As for the fault propagation of power system, the fault types decide the states of each node, the probability of being affected and further propagation. The dynamic characteristics of power systems are dependent on the characteristics of each node, the type of fault, and the system recovery ability.

III. FAULT SEVERITY INDEX OF NODE

Existing research generally ignores the node interaction during fault propagation, while this paper holds that the fault severity is the dynamic cause of state transformation in epidemic model, and quantifies it from the perspective of node frequency, power flow and voltage index.

In power system, the fault of a node will spread with a certain probability. If the immunity of adjacent nodes is strong,

the probability decreases to zero. This paper revolves around the influence from the state of nodes to fault propagation and proposes a feasible method to characterize the severity of node fault propagation.

A. Frequency Response Index of Node

Inertia of synchronous generator rotor or turbine is an important parameter that affects frequency response in power system. This part proposes a node frequency response index, which quantifies the anti-disturbance ability of the node to sustain frequency change with the use of two variables, generator inertia and relative node position.

When a power disturbance occurs in the system, the power disturbance magnitude, inertia time constant and frequency satisfy the rocking equation:

$$2H_{\text{sys}}\frac{\mathrm{d}f_{\text{coi}}}{\mathrm{d}t} = \Delta P \tag{3}$$

where f_{coi} and ΔP represent the inertia center frequency and power unbalance measure of the system, respectively. If the inertia of the system, H_{sys}, decreases but the frequency disturbance keeps same, the frequency change rate will increase, while the system frequency stability and the node's anti-disturbance ability will decline.

In fact, when a power disturbance occurs at different nodes, the frequency responses are different. With reference to the response of the equivalent inertia of the system to the change of system power and frequency, the calculated inertia of the node is defined as the ratio of the power disturbance magnitude to the initial frequency change rate of the node after power disturbance:

$$H_{ck} = \frac{\Delta P}{2 \cdot \mathrm{d}f_k/\mathrm{d}t} \tag{4}$$

where H_{ck}, ΔP and f_k are, respectively, the calculated inertia, the disturbance power and the frequency of the node k. The calculated inertia of the node k is numerically equivalent to the inertia time constant of a virtual synchronous generator at the node k. If node k is influenced by power disturbance and the load response to unbalanced power is ignored, the unbalanced power will be distributed to each synchronous generator node according to the synchronization power coefficient. The synchronization power coefficient [30] is as follows:

$$D_{jk} = V_j V_k B_{jk} \cos\delta_{jk0} \tag{5}$$

where D_{jk} is the synchronous power coefficient of the generator. V_j and V_k are the internal potential amplitude of the generator and the voltage amplitude of the node. B_{jk} is the susceptance assigned to the internal potential node of the generator and the system admittance matrix after failure. δ_{jk0} is the initial phase angle difference of the voltage between the generator and the node. Since the renewable energy generator does not have the rotor motion characteristics and electromagnetic transient characteristics of the synchronous generator set, it cannot share the unbalanced power of the system when fault occurs.

Assume that there are n generator nodes and m other nodes in the power system, and the admittance matrix is:

$$\boldsymbol{Y} = \begin{bmatrix} \boldsymbol{Y}_{nn} & \boldsymbol{Y}_{nm} \\ \boldsymbol{Y}_{mn} & \boldsymbol{Y}_{mm} \end{bmatrix} \tag{6}$$

where $\boldsymbol{Y}_{nn}$ is the self-admittance of generator node while $\boldsymbol{Y}_{mm}$ is that of other nodes. $\boldsymbol{Y}_{nm}$ and $\boldsymbol{Y}_{mn}$ are the mutual admittance between them. If we take into consideration the generator transient reactance, and simplify the load and energy to equivalent admittance [31], the augmented admittance matrix of power system is:

$$\boldsymbol{Y}_s = \begin{bmatrix} \boldsymbol{Y}_n & -\boldsymbol{Y}_n & \boldsymbol{0} \\ -\boldsymbol{Y}_n & \boldsymbol{Y}_{nn} + \boldsymbol{Y}_n + \boldsymbol{Y}_{ln} & \boldsymbol{Y}_{nm} \\ \boldsymbol{0} & \boldsymbol{Y}_{mn} & \boldsymbol{Y}_{mm} + \boldsymbol{Y}_{lm} \end{bmatrix} \tag{7}$$

where $\boldsymbol{Y}_s$ is the augmented matrix of system, while $\boldsymbol{Y}_n$ is the diagonal matrix formed by generator transient reactance. $\boldsymbol{Y}_{ln}$ and $\boldsymbol{Y}_{lm}$ are respectively the equivalent load admittance of energy node and that of other nodes.

The above augmented matrix is expressed as:

$$\boldsymbol{Y}_s = \begin{bmatrix} \boldsymbol{Y}_1 & \boldsymbol{Y}_2 \\ \boldsymbol{Y}_3 & \boldsymbol{Y}_4 \end{bmatrix} \tag{8}$$

where $\boldsymbol{Y}_1 = \boldsymbol{Y}_n$, $\boldsymbol{Y}_2 = [-\boldsymbol{Y}_n \quad \boldsymbol{0}]$, $\boldsymbol{Y}_3 = [-\boldsymbol{Y}_n \quad \boldsymbol{0}]^{\mathrm{T}}$, $\boldsymbol{Y}_4 = \begin{bmatrix} \boldsymbol{Y}_{nn} + \boldsymbol{Y}_n + \boldsymbol{Y}_{ln} & \boldsymbol{Y}_{nm} \\ \boldsymbol{Y}_{mn} & \boldsymbol{Y}_{mm} + \boldsymbol{Y}_{lm} \end{bmatrix}$.

The corresponding network equation is:

$$\begin{bmatrix} \boldsymbol{I}_n \\ \boldsymbol{0} \end{bmatrix} = \boldsymbol{Y}_s \begin{bmatrix} \boldsymbol{U}_e \\ \boldsymbol{U}_{\text{net}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{Y}_1 & \boldsymbol{Y}_2 \\ \boldsymbol{Y}_3 & \boldsymbol{Y}_4 \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_e \\ \boldsymbol{U}_{\text{net}} \end{bmatrix} \tag{9}$$

The injected current of network node is ignored here. $\boldsymbol{I}_n$ represents the system current vector injected by generator, $\boldsymbol{U}_e$ and $\boldsymbol{U}_{\text{net}}$ are the voltage vectors of generator potential nodes and those of network nodes.

The voltage relationship between the network node and the generator node is expressed as:

$$\boldsymbol{Y}_3\boldsymbol{U}_e + \boldsymbol{Y}_4\boldsymbol{U}_{\text{net}} = \boldsymbol{0} \tag{10}$$

The voltage of the network node expressed by the generator voltage is:

$$\boldsymbol{U}_{\text{net}} = -\boldsymbol{Y}_4^{-1}\boldsymbol{Y}_3\boldsymbol{U}_e = \boldsymbol{R}\boldsymbol{U}_e \tag{11}$$

According to the simplification operation, the frequency of the node k can be expressed as follows [32]:

$$f_k = \frac{\sum_{j\in k} r_{kj} V_j f_j}{\sum_{j\in k} r_{kj} V_j} \tag{12}$$

where f_j and V_j are the frequency and the voltage value of the generator j. r_{kj} is the element of the matrix $\boldsymbol{R}$ which corresponds to the frequency relationship between the node k and the generator j.

We can get the matrix $\boldsymbol{Y}_s'$ by the rearrangement of $\boldsymbol{Y}_s$, where $n+1$ row (column) is the nodes of internal voltage for generators and the fault nodes, while the remaining row (column) is residual generator nodes and load nodes. Equation (9) can be rewritten as:

$$\begin{bmatrix} \boldsymbol{I}_{\text{EF}} \\ \boldsymbol{0} \end{bmatrix} = \boldsymbol{Y}_s' \begin{bmatrix} \boldsymbol{U}_{\text{EF}} \\ \boldsymbol{U}_{\text{R}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{Y}_1' & \boldsymbol{Y}_2' \\ \boldsymbol{Y}_3' & \boldsymbol{Y}_4' \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{\text{EF}} \\ \boldsymbol{U}_{\text{R}} \end{bmatrix} \tag{13}$$

where $\boldsymbol{U}_{\text{EF}}$ is the generator internal voltage and fault node voltage. $\boldsymbol{U}_{\text{R}}$ is the voltage of residual generator nodes and

load nodes. $\boldsymbol{I}_{\mathrm{EF}}$ is the injected current of generator potential nodes and fault nodes. So, it can be rewritten in the equation below as:

$$\boldsymbol{I}_{\mathrm{EF}} = \boldsymbol{Y}_{\mathrm{EF}}\boldsymbol{U}_{\mathrm{EF}} = (\boldsymbol{Y}_1' - \boldsymbol{Y}_2'\boldsymbol{Y}_4'^{-1}\boldsymbol{Y}_3')\boldsymbol{U}_{\mathrm{EF}} \tag{14}$$

where B_{iF} is the susceptance of system admittance matrix after shrinking to the generator potential node and the fault node, which can be obtained from $\boldsymbol{Y}_{\mathrm{EF}}$.

When there is power disturbance ΔP at the node k, the unbalanced power is distributed to each generator node according to the synchronous power coefficient. The unbalanced power picked up by the generator j can be written as:

$$\Delta P_j = \frac{D_{jk}}{\sum_{j\in k} D_{jk}}\Delta P \tag{15}$$

where D_{jk} and ΔP_j are the corresponding synchronous power coefficient and the picked unbalanced power of the generator respectively.

With the combination of the generator swing equation and Equation (15), the frequency change rate of the generator can be obtained:

$$\frac{\mathrm{d}f_j}{\mathrm{d}_t} = \frac{D_{jk}\Delta P}{2H_j \cdot \sum_{j\in k} D_{jk}} \tag{16}$$

Substitute (12) and (13) into (4), and calculated inertia of the node can be expressed as:

$$H_{ck} = \frac{\sum_{j\in k} D_{jk}}{\sum_{j\in k}\left(\frac{1}{H_j} r_{kj} \cdot D_{jk}\right)} \tag{17}$$

The major concern is the influence of susceptance. Simultaneously, with the consideration that the system voltage is near the rated value, the power allocation coefficient is simplified, and the calculated inertia of node k is finally obtained as follows:

$$H_{ck} = \frac{\sum_{j\in k} B_{jk}}{\sum_{j\in k}\left(\frac{1}{H_j} r_{kj} \cdot B_{jk}\right)} \tag{18}$$

The larger the calculated inertia is, the stronger the ability to resist fault disturbance becomes. So, the node frequency response index $D(v_i)$ is defined as:

$$D(v_i) = \frac{1}{H_{ci}} = \frac{\sum_{j\in i}\left(\frac{1}{H_j} r_{ij} \cdot B_{ji}\right)}{\sum_{j\in i} B_{ji}} \tag{19}$$

where H_j is inertia time constant of the generator j. This index comprehensively considers the inertia of each generator and the distance between the fault node and each generator. It can reflect the frequency response ability of each node after the system failures. The frequency response index considers the low inertia of the renewable energy node and the inertia distribution of the system, which can truly reflect the operation condition of the power system with high penetration of renewable energy. In this paper, the frequency response index is considered as an inherent attribute of the network node to deal with the fault and remains unchanged during the fault.

B. Power Flow Coupling Index of Node

When a node of power system fails, it first propagates the fault to adjacent nodes, and then uses the coupling path to affect other nodes. When the fault propagates in the power system, a node can become an affected node, and may also affect other nodes. This paper regards the nodes with strong coupling relationship with other nodes as vulnerable nodes. The coupling entropy of source and terminal nodes are defined to represent the degree of interdependence between a node and its neighbors.

With reference to [33], this part divides nodes into source nodes and terminal nodes. Source nodes provide power to their neighbors while terminal nodes extract power from source nodes. Therefore, the magnitude of node input power and output power is defined as the dynamic coupling index between the node and its adjacent nodes.

1) Coupling Dependency Utility of Source Nodes

As shown in Fig. 2, the direction of edge indicates the direction of power input and output between nodes.

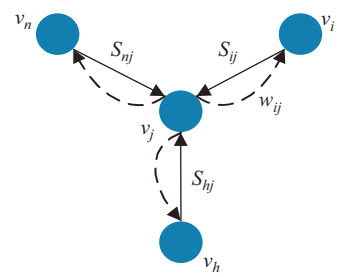

Fig. 2. Coupling model of source nodes.

The direction of interdependence between nodes is opposite. Weight w indicates dependency degree between neighboring nodes, and w_{ij} refers to the degree to which node v_i is affected by v_j. Since the effectiveness of v_j depends on v_i, v_n and v_h, which will affect its neighbor source nodes v_i, v_n and v_h, the edge weight is:

$$w_{ij} = \frac{S_{ij}/\sum_{n\in\lambda_j^{\mathrm{in}}} S_j}{\sum_{j\in\lambda_i^{\mathrm{out}}}\left(S_{ij}/\sum_{n\in\lambda_j^{\mathrm{in}}} S_j\right)} \tag{20}$$

where S_{ij} is the apparent power transmitted from v_i to v_j. λ_j^{in} is the set of nodes that input power to v_j, while λ_i^{out} is the set of nodes that extract power from v_i. The numerator of (20) represents the ratio of the power absorbed from v_i by v_j to the total power absorbed by v_j. The denominator represents the ratio of output power from node v_i to node v_j to the total output power from node v_i. Therefore, influence of terminal nodes on source nodes can be expressed by w_{ij}. With the consideration of the connection between v_i and other nodes, coupling entropy can be used to define the interdependence utility when v_i is affected by neighbor nodes. Set p_{ij} to represent normalized edge weight:

$$p_{ij} = \frac{w_{ij}}{\sum_{j\in\lambda_i^{\mathrm{out}}} w_{ij}} \tag{21}$$

Because $\sum p_{ij} = 1$ the coupling entropy L_i^{out} of v_i can be defined as:

$$L_i^{\mathrm{out}} = \left(1 - \sum p_{ij}\ln p_{ij}\right)\sum w_{ij} \tag{22}$$

2) Coupling Dependency Utility of Terminal Nodes

As shown in Fig. 3, node v_i extracts the power S_{mi}, S_{li} and S_{gi} from v_m, v_l and v_g, respectively, which indicates the dependence of v_i on v_m, v_l and v_g. That is to say, v_i will fail when v_m, v_l and v_g fail. Therefore, the edge weight w_{mi} which reflects the dependency utility of v_i on v_m is:

$$w_{mi} = \frac{S_{mi}}{\sum_{m \in \lambda_i^{\text{in}}} S_{mi}} \tag{23}$$

where λ_i^{in} is the set that outputs power to v_i. The edge weight w_{mi} can also be obtained in the same way above.

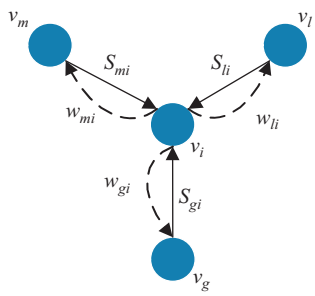

Fig. 3. Coupling model of terminal nodes.

Therefore, the normalized edge weight set p_{mi} is expressed as:

$$p_{mi} = \frac{w_{mi}}{\sum_{m \in \lambda_i^{\text{in}}} w_{mi}} \tag{24}$$

The coupling entropy L_i^{in} of terminal node v_i represents the dependency utility of v_i on its adjacent source nodes, which can be expressed as:

$$L_i^{\text{in}} = \left(1 - \sum p_{mi} \ln p_{mi}\right) \sum w_{mi} \tag{25}$$

According to the definitions of L_i^{out} and L_i^{in}, the coupling relationship between nodes and their neighbors is defined as the node power flow coupling index $L(v_i)$:

$$L(v_i) = \rho L_i^{\text{out}} + (1 - \rho) L_i^{\text{in}} \tag{26}$$

where ρ is the distribution coefficient of dependent utility, and $\rho \in [0, 1]$. The assumption in this paper is that L_i^{in} is as important as L_i^{out} and $\rho = 0.5$. The influence of a node fault on other nodes along the coupling path in the system is analyzed through the iterative process of node power flow index in the system.

3) Voltage Offset Index of Node

In the peak load period, when load significantly increases to the maximum transmission power, the voltage will slump. This will further lead to voltage collapse and cause potential cascading failures. On the contrary, when input power is greater than output, the voltage will rise. Therefore, the voltage offset of node can reflect real-time operation status of node. The voltage offset index of node can be defined as:

$$I(v_i) = \left| \frac{V_t(v_i) - V_{cr}(v_i)}{V_0(v_i) - V_{cr}(v_i)} \right| \tag{27}$$

where $V_t(v_i)$, $V_0(v_i)$, $V_{cr}(v_i)$ are the actual value of node voltage, the initial voltage value of the node and the critical voltage value of the node. $I(v_i)$ can accurately distinguish the severity of voltage instability of each node in the process of fault development.

4) Fault Severity Index based Game AHP Entropy Weight Method

In the actual evaluation work, the evaluation index should not only fully extract the information contained in the objective data, but also fully respect the subjective opinions of experts, which requires the utility risk entropy weight and the expert weight to be combined in a reasonable way. Game AHP entropy weight method can effectively solve this problem [34]. The basic idea of this method is to find agreement or compromise among different weights, so that the deviation between the ideal comprehensive weight and the subjective and objective weights is minimized, and the information of each subjective and objective weight value is retained as much as possible.

The fault severity is related to frequency stability, power flow coupling characteristics and voltage offset degree [35]. Based on the factors above, the fault severity of node is defined as:

$$R(v_i) = \omega_1 \times D(v_i) + \omega_2 \times L(v_i) + \omega_3 \times I(v_i) \tag{28}$$

where $R(v_i)$ is the fault severity of node, and ω_1, ω_2, ω_3 are the weight coefficients. $D(v_i)$, $L(v_i)$ and $I(v_i)$ are all normalized indicators here. Game AHP entropy weight method is used to obtain the comprehensive weight of each evaluation index, and the steps are as follows:

Step 1: Analytic hierarchy process and entropy weight method in [36] are used to calculate subjective weight $\boldsymbol{W}_1$ and objective weight $\boldsymbol{W}_2$.

Step 2: The index combination weight vector $\boldsymbol{W}$ expressed by the linear combination of vectors $\boldsymbol{W}_1$ and $\boldsymbol{W}_2$ is established as:

$$\begin{aligned} \boldsymbol{W} &= \begin{pmatrix} \lambda_1 w_{11} + \lambda_2 w_{12} \\ \lambda_1 w_{12} + \lambda_2 w_{22} \\ \lambda_1 w_{13} + \lambda_2 w_{23} \end{pmatrix} = \begin{pmatrix} w_{11} & w_{21} \\ w_{12} & w_{22} \\ w_{13} & w_{23} \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} \\ &= \lambda_1 \boldsymbol{W}_1 + \lambda_2 \boldsymbol{W}_2 \end{aligned} \tag{29}$$

where λ_1 and λ_2 are linear combination coefficients.

Step 3: The objective function is established according to the game theory. The aim is to minimize the sum of deviation between index combination weight $\boldsymbol{W}$ and $\boldsymbol{W}_1$ plus $\boldsymbol{W}_2$, and to seek the optimal linear combination coefficient λ_1 and λ_2. This weight is the optimal combination weight $\boldsymbol{W}$. The objective function is:

$$\begin{aligned} &\min \left\| \sum_{i=1}^{2} \lambda_i \boldsymbol{W}_i^{\mathrm{T}} - \boldsymbol{W}^{\mathrm{T}} \right\|_2 \\ &\lambda_1 > 0; \lambda_2 > 0. \end{aligned} \tag{30}$$

Step 4: According to the differential theorem, the condition for obtaining the first derivative of minimum value of (30) is:

$$\begin{cases} \lambda_1 \boldsymbol{W}_1^{\mathrm{T}} \boldsymbol{W}_1 + \lambda_2 \boldsymbol{W}_1^{\mathrm{T}} \boldsymbol{W}_2 = \boldsymbol{W}_1^{\mathrm{T}} \boldsymbol{W}_1 \\ \lambda_1 \boldsymbol{W}_2^{\mathrm{T}} \boldsymbol{W}_1 + \lambda_2 \boldsymbol{W}_2^{\mathrm{T}} \boldsymbol{W}_2 = \boldsymbol{W}_2^{\mathrm{T}} \boldsymbol{W}_2 \end{cases} \tag{31}$$

Thus, the values of λ_1 and λ_2 can be calculated.

Step 5: The linear combination coefficients are normalized.

$$\begin{cases} \lambda_1^* = \dfrac{|\lambda_1|}{|\lambda_1| + |\lambda_2|} \\ \lambda_2^* = \dfrac{|\lambda_2|}{|\lambda_1| + |\lambda_2|} \end{cases} \tag{32}$$

The optimal combination weight of evaluation index is:

$$\boldsymbol{W}^* = \lambda_1^* \boldsymbol{W}_1 + \lambda_2^* \boldsymbol{W}_2 \tag{33}$$

According to the analysis above, the fault severity index $R(v_i)$ of each node can be determined by combining the comprehensive weight with the normalized indices. The influence degree of each node by the fault disturbance can be evaluated according to the index.

IV. Analysis of Fault Propagation Model

The key to the power system fault propagation model based on epidemic model is to determine the infection rate and cure rate. The model considers both of the complex power network topology and the relationship between internal and external data of the power grid. The analysis steps are shown in Fig. 4.

First, we calculate the frequency response index based on the inertia distribution of the generator and the system network structure. The power flow coupling index is obtained by source node coupling and terminal node coupling. The voltage offset index is calculated from the perspective of voltage instability. In addition, to determine the node fault severity index, we utilize the frequency response index, power flow coupling index, voltage offset index, and the game AHP entropy weight method. By setting the threshold of the change rate of the node fault severity index, changes of node states can be detected.

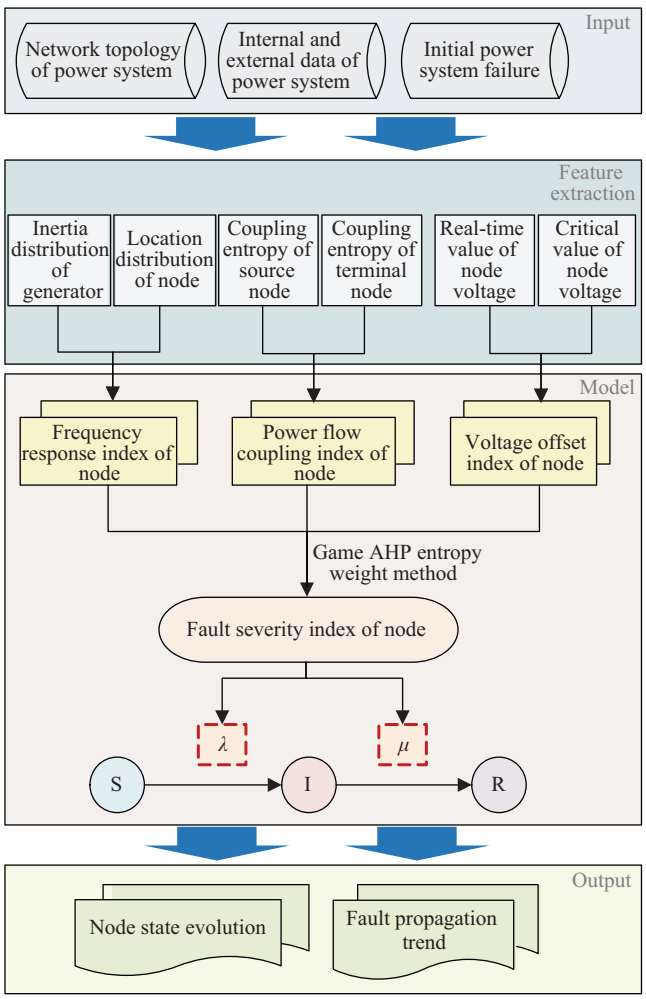

Fig. 4. Analysis steps of power system fault propagation.

The random SIR epidemic model [29] is used to explain how the SIR epidemic model simulates the fault propagation process. To facilitate analysis, it can be set $\{S(t), I(t), t \geq 0\}$ as the SIR stochastic model of a binary random process, and then has the following joint probability function:

$$P_{i,j}(t) = P\{S(t) = i, I(t) = j\} \tag{34}$$

Suppose that from time t, in infinitesimal $\mathrm{d}t$ time, there are only three cases in a node in the state environment:

1) The infection probability of a single vulnerable node is $\lambda S(t)I(t)\mathrm{d}t$, and the number of fault nodes increases by one;

2) The probability that a single fault node is recovered is $\mu I(t)\mathrm{d}t$, and the number of fault nodes decreases by one;

3) The probability that the state does not change is $1 - (\lambda S(t) + \mu)I(t)\mathrm{d}t$, and the number of nodes in each state is unchanged.

Therefore, the transition probability of the binary stochastic process is as follows:

$$\begin{aligned} &P\{(S,I)_{t+\mathrm{d}t} = (i,j) | (S,I)_t = (m,n)\} \\ &= \begin{cases} \lambda S(t)I(t)\mathrm{d}t + o(\mathrm{d}t), & (i,j) = (m-1, n+1) \\ \mu I(t)\mathrm{d}t + o(\mathrm{d}t), & (i,j) = (m, n-1) \\ 1 - (\lambda S(t) + \mu)I(t)\mathrm{d}t + o(\mathrm{d}t), & (i,j) = (m,n) \end{cases} \end{aligned} \tag{35}$$

where $o(\mathrm{d}t)$ represents the high-order infinitesimal quantity of $\mathrm{d}t$.

In the random SIR model, $\{S(t), I(t), t \geq 0\}$ can be analyzed separately as follows:

1) Only the state transition of the vulnerable node $S(t)$ is studied, and $S(t)$ is "only out but not in". It can be known that the single $\{S(t), t \geq 0\}$ is merely an extinction process. When $\mathrm{d}t$ is sufficiently small, the transition probability of the process is as follows:

$$\begin{aligned} &P\{S_{t+o(t)} = j | S_t = i\} \\ &= \begin{cases} \lambda S(t)I(t)\mathrm{d}t + o(\mathrm{d}t), & j = i-1 \\ 1 - \lambda S(t)I(t)\mathrm{d}t + o(\mathrm{d}t), & j = i \\ o(\mathrm{d}t), & \text{else} \end{cases} \end{aligned} \tag{36}$$

2) Only the state transition of the fault node $I(t)$ is studied, and $I(t)$ is "in and out", which is known to be a "birth and death process". When $\mathrm{d}t$ is sufficiently small, the transition probability of the process is as follows:

$$\begin{aligned} &P\{I_{t+\mathrm{d}t} = j | I_t = i\} \\ &= \begin{cases} \lambda S(t)I(t)\mathrm{d}t + o(\mathrm{d}t), & j = i+1 \\ \mu I(t)\mathrm{d}t + o(\mathrm{d}t), & j = i-1 \\ 1 - (\lambda S(t) + \mu)I(t)\mathrm{d}t + o(\mathrm{d}t), & j = i \\ o(\mathrm{d}t), & \text{else} \end{cases} \end{aligned} \tag{37}$$

3) Only state transition of the recovered node $R(t)$ is studied, and $R(t)$ is a "pure generative process". When $\mathrm{d}t$ is sufficiently small, the transition probability of the process is as follows:

$$P\{R_{t+\mathrm{d}t}=j|R_t=i\} = \begin{cases} \mu I(t)\mathrm{d}t+o(\mathrm{d}t), & j=i+1 \\ 1-\mu I(t)\mathrm{d}t+o(\mathrm{d}t), & j=i \\ o(\mathrm{d}t), & \text{else} \end{cases} \tag{38}$$

The infection rate λ and cure rate μ can be obtained by the square method [37], thus determining the rules in the dynamic model of fault propagation. Finally, through the analysis of power grid fault propagation, the fault node can be located, and the development of cascading failures can be predicted, which can provide assistance to relevant personnel.

Compared with conventional cascading failure prediction methods, the fault propagation model based on SIR epidemic model can provide a quantitative evaluation of each node in different stages of cascading failure development and perceive the overall safety level of the power system. Although traditional SI and SIS epidemic models can simulate the cascading failure process, they cannot reflect the recovery process after the failure or reflect the real fault situation. The state transition parameters of traditional SIR epidemic model lack a theoretical basis. It is believed that the set parameters are very random, which could incur errors and give rise to a large gap between the predicted and actual values. According to the characteristics of fault propagation mode in real-time power systems, this paper researches the importance of each influencing factor during fault propagation and diffusion. The evolution equations of nodes in different states are established to simulate process of fault propagation and diffusion. The influence weight of nodes in different states is obtained through repeated calculations, which is used as the quantitative value of the system infection rate and cure rate to predict potential fault propagation trend.

V. Case Study

A. Simulation System and Cascading Failure Setting

In order to verify validity of the proposed model, this paper analyzes the fault propagation process on IEEE 39-bus system in PSCAD/EMTDC, and the simulation step size is 20 μs. As shown in Fig. 5, there are 10 generator nodes, 12 transformer nodes, 17 load nodes, 39 buses and 46 lines, of which node 30 and node 31 are replaced by wind farms with equal capacity. The wind power penetration rate is 20%. The CPU model of the computer used in the simulation is Intel-i5-12400-2.5 GHz, and the memory is 8 GB.

Nodes 15, 20 and 23 are industrial loads. The active power of the three nodes suddenly changes to 448 MW, 879.2 MW and 346.5 MW (1.4 times of the original power) at 40 s. The load of other nodes remains unchanged, and the system load increases by 478.2 MW. According to historical data, the single-phase ground fault is most likely to happen, and the highest probability is about 65% [38], so this paper chooses this kind of fault as the simulation object. We assume that branches 14–15 and 16–19 had transient single-phase ground fault due to overload at 50 s, and the fault is cleared after 0.5 s. At 55 s, the active power of the three nodes recovers to 320 MW, 628 MW and 247.5 MW, and the system operation status remains stable.

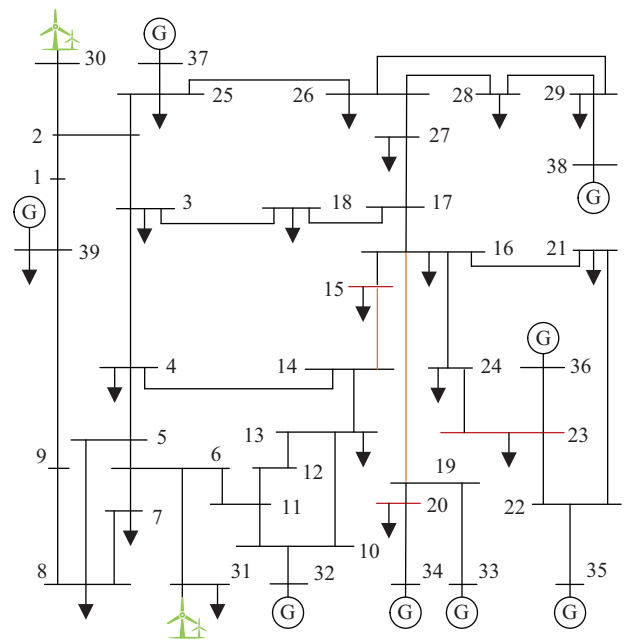

Fig. 5. Structure of IEEE 39-bus system.

B. Analysis of Cascading Failure Process

The voltage offset of each node during cascading failures is analyzed with the load node voltage as an example. The voltage curves of load nodes are shown in Fig. 6.

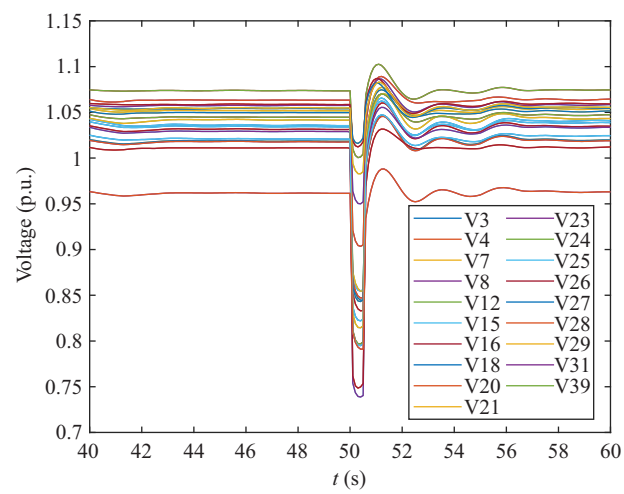

Fig. 6. Voltage curves of load nodes.

From 40 s to 50 s, with the sudden power increase of nodes 15, 20 and 23, the voltage of load nodes drops and reaches the lowest at 41.3 s. Then there is a gradual increment on the voltage and slight oscillation near the stable value at 43.6 s, which is still lower than the initial. During the process, the load disturbance has little impact on the voltage offset, and

the maximum voltage offset is 1.3%. When $t = 50$ s, due to single-phase ground fault on branches 14–15 and 16–19, the voltage of load node slumps. The maximum voltage deviation 30.2% occurs in node 15 at 50.4 s. After the fault is cleared at 50.5 s, the node voltage soars and exceeds the initial voltage. The maximum voltage offset 2.9% occurs in node 26 at 51 s. From 51 s to 55 s, the load node voltage drops to a stable value and presents a small oscillation, which is lower than the initial. As the active power of nodes 15, 20 and 23 recovers to the original state after 55 s, the load node voltage rises and oscillates slightly near the stable value, which is approximately equal to the initial.

Next is the analysis of the power change of each node during cascading failure. The specific analysis object is the branch power connected with nodes 15, 20 and 23. The branch power curve is shown in Fig. 7.

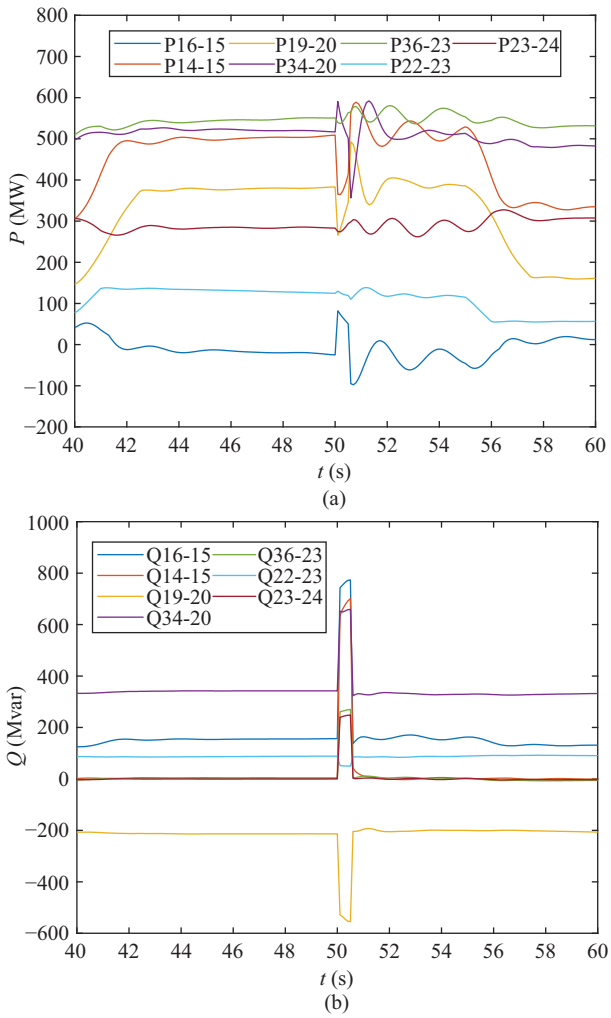

Fig. 7. Power curve of branches. (a) Active power curve of branches. (b) Reactive power curve of branches.

From 40 s to 50 s, as the 478.2 MW active load increases, the added load will be shared by each generator, so the active power of branches will be disturbed but the active load fluctuation will cause little reactive power disturbance of branches. During 50 s to 50.5 s, the voltage drops because the single-phase ground fault of branches 14–15 and 16–19 causes a redistribution of system active power. Consequently, active power of each branch changes, while the generators increase excitation current and send out more reactive power to maintain the node voltage. The reactive power of each branch presents a wide scale of oscillation. Due to the short fault duration, the system maintains transient stability. From 50.5 s to 55 s, the active power oscillates slightly near a certain stable value, which is greater than the initial, and the change of reactive power of branches is small. After 55 s, the active load returns to the original state and the power of each branch returns to the initial value after a short disturbance. Then the whole system returns to the initial stable state.

C. Calculation Results and Analysis

Take the indices of 55.5 s as an example. The frequency response index, power flow coupling index and voltage offset index of each node can be calculated through the analysis in Section III. After the normalization of all index vectors, the objective weight of index is $[0.3409, 0.4610, 0.1981]^{\mathrm{T}}$ and the subjective weight is $[0.2269, 0.5407, 0.2324]^{\mathrm{T}}$. If we substitute data into (28)–(33), the fault severity of each node at $t = 55.5$ s is obtained, and the calculation results are shown in Fig. 8.

At the fault development stage, what distinguishes infected nodes from others is that their fault severity index change rate is greater than 0.482, and the average level of nodes 15, 20 and 23 at 40 s. At fault recovery stage, when the change rate of fault severity of infected node is less than average change rate of nodes 15, 20 and 23 at 40 s, this node changes into a recovered node. When $t = 55.5$ s, there are 9 infected and 30 cured nodes and the spread of cascading failures gradually slows down.

Take node 15 as an example. The changes in frequency response index, power flow coupling index, voltage offset index and fault severity index during the actual failure process are shown in Fig. 9. The active power of node 15 increases by 128 MW at 40 s and the change rate of fault severity is 0.531, which is greater than average change rate of fault severity index 0.482 of nodes 15, 20 and 23 at 40 s. Therefore, node 15 changes from a susceptible state to an infected state at 40 s. During the recovery stage, the change rate of fault severity of node 15 is 0.283 at 50.9 s, which for the first time decreases to a level below 0.482 after 50.5 s. That is to say, node 15 changes from an infected state to a recovered state at 51.1 s.

During the failure development stage, the continuous time calculation of the system is carried out and the computational time is 5.11 s. The comparison of infected nodes density between the actual situation and prediction is shown in Fig. 10.

The Maximum Infected Density Peak (MIDP) is the maximum value of infected density in the process of failure propagation in the area. When the infected density reaches the maximum value, the system is the most vulnerable and the cascading failure propagation process reaches the most dangerous degree, so we take MIDP as the index to evaluate the maximum degree of cascading failure propagation. It can be seen from Fig. 10 that the predicted fault propagation trend in this paper is close to the real trend and MIDP arrives earlier than the actual peak. The risk of cascading failure can

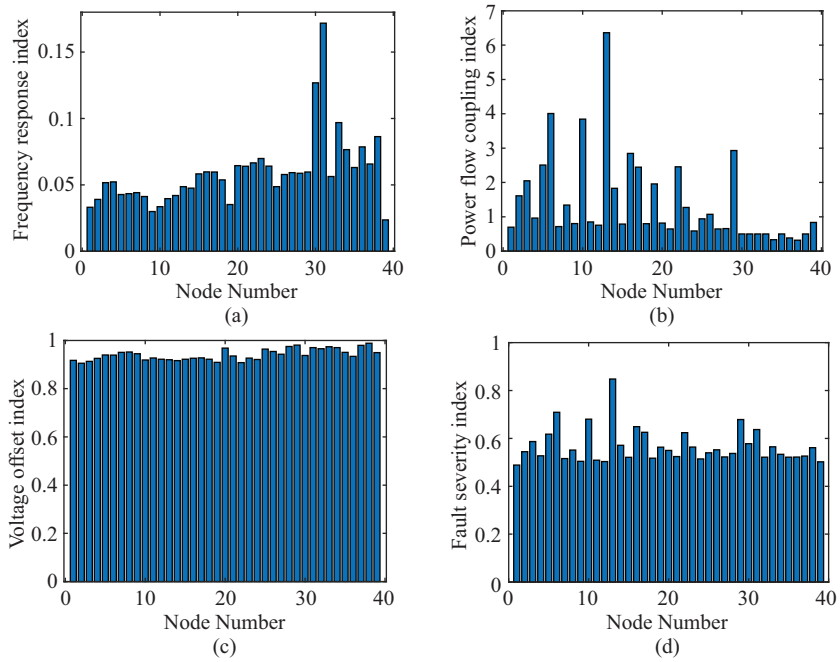

Fig. 8. Calculation of four indices at 55.5 s. (a) Frequency response index at 55.5 s. (b) Power flow coupling index at 55.5 s. (c) Voltage offset index at 55.5 s. (d) Fault severity index of each node at 55.5 s.

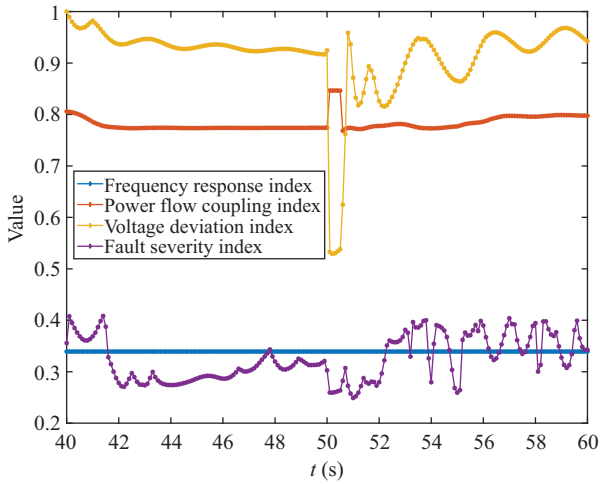

Fig. 9. Change of four indices during the failure process.

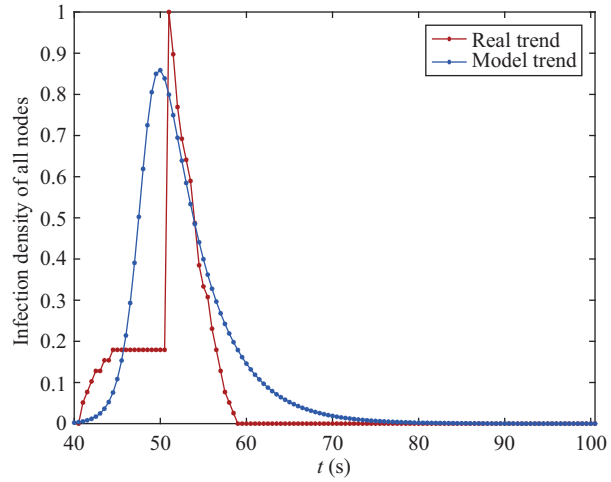

Fig. 10. Infected nodes density of actual and predicted value in IEEE 39-bus system.

be effectively reduced by setting a reasonable threshold for the infected density and adopting corresponding containment control strategies before further spread.

In order to verify the effectiveness of method proposed on the development situation of cascading failures, the threshold of starting the control and protection strategy is defined by the quantile of the sufficiency function [39] in power system. This paper sets the system security measures will act when the infection density of system nodes is detected to be over 0.2 and the emergency control strategy is referred to [40]. The maximum adjustable amount of each control node in the system is set at 50% of the normal value of the node (without considering the added output adjustment of renewable generation nodes). The initial fault is set as the power flow of 191.94 MW, 585.53 MW and 431.94 MW, respectively, on branches 1–2, 2–3 and 26–27. At this time, the three branches are all overloaded. After four rounds of overload control, the following control strategy is formed: the output reduction of nodes 30, 37 and 38 is 80.65 MW; the output reduction of node 33 is 57.32 MW, the output increase of node 32 is 57.32 MW, and the cutting load of nodes 3, 8 and 27 is 80.65 MW. The transmission power of branches 1–2, 2–3 and 26–27 is restored to 160.85 MW, 467.74 MW and 338.87 MW, respectively, and the branch is below the maximum transmission power. The MIDP of the whole process is calculated to be 0.5487. It is significantly less than the MIDP value (0.8718), which receives no security measures. The indicators proposed for the power systems with high-penetration renewable energy can effectively detect the development trend of cascading

failures and thus provide effective theoretical support for the identification of key blocking points of the power system and potential fault control strategies.

D. Method Comparison Analysis

1) Effectiveness Analysis of the Method

In order to verify the feasibility of this method, the control group test is designed based on WSCC3 9-bus system with reference to [41]. The calculation results of the two methods under different disturbances are in Table II.

TABLE II
COMPARISON OF THE RESULTS BETWEEN TWO METHODS

Disturbance number	Perturbation	MIDP	The real part variance of track characteristic root
1	Steady state operation without disturbance	0	6.84×10^{-5}
2	Node 5 increases the load by 20% for 0.2 s	0.1837	5.318
3	Node 5 increases the load by 40% for 0.2 s	0.3722	7.915
4	Node 5 increases the load by 80% for 0.2 s	0.6906	8.953

It can be seen from Table II that MIDP and the real part variance of track characteristic root both increase with the continuous improvement of disturbance, which can reflect the severity of disturbance. These indicators prove that the model established in this paper can be used for fault detection. But there are many steps converting nonlinearity into linearization in methods of [41], which cannot meet the requirements of online evaluation in terms of rapidity. To form a contrast, the method in this paper can complete the calculation only with the real-time power flow and voltage of each node, which can meet the time requirements of real-time evaluation.

2) Advantages Analysis of the Method

In order to further illustrate the advantages of SIR epidemic model, additional cases are supplemented.

For comparative analysis, the proposed method, the traditional method of regional safety evaluation index [42] and the method based on SI epidemic model are used to describe the severity of cascading failures of the system. Other conditions remain unchanged, and the comparison of the three methods on the spreading degree of cascading failures is shown in Fig. 11.

Taking 20% fault severity as the action threshold of control and protection strategy, it can be seen from Fig. 11 that for the method based on SIR epidemic model, the development of cascading failures reaches the dangerous level at 46 s, while for the regional security assessment index, the development of cascading failures reaches the dangerous level at 49.4 s. As for the results of cascading failures situation detection, the method proposed in this paper can perceive the development trend of cascading failures in advance. Based on the traditional electricity quantity index, it pays more attention to the fault severity under the current operating state. SI epidemic model believes that there is no cure for nodes in the fault propagation of power grid, which is obviously inconsistent with the actual fault propagation process. In addition, SI epidemic model

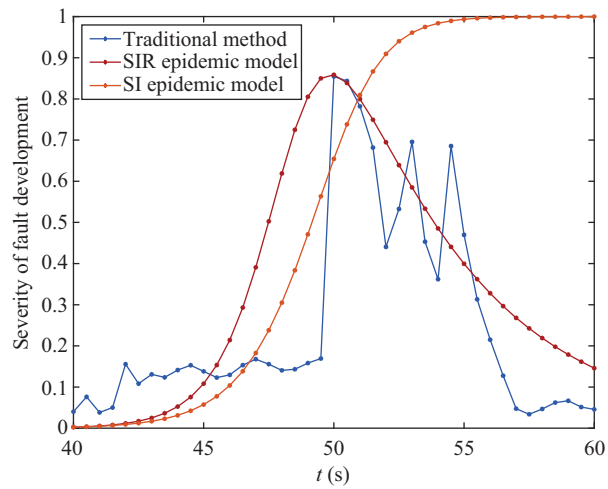

Fig. 11. Severity of fault development in three methods.

cannot be used to verify the control and protection strategy or to simulate the cure process. SIR epidemic model considers the fault recovery process, and by introducing the cure rate index, it can verify the intervention degree of the control and protection strategy on the cascading failure, thus closer to the actual fault propagation process. Therefore, the detection method of cascading failure proposed in this paper can realize the overall judgment in advance. If the corresponding control and protection strategies are taken after the perceived cascading failure risk reaches the set threshold, the spreading process of cascading failures can be effectively stopped, the safe operation level of the system can be improved, and the occurrence of large power outages can be avoided as much as possible.

E. Case Comparison Analysis

1) Comparison of Different Infection and Cure Rates

Based on fault propagation model, the fault propagation width is analyzed in Fig. 12 by fixing the infection rate and cure rate. We set the step size δ at 0.05 and the maximum coverage during fault propagation is analyzed within the range (0, 1).

It can be found from Fig. 12 that there is a positive correlation between MIDP and infection rate and a negative one between MIDP and cure rate in all six situations. The change of MIDP indicates that the fault propagation is different in unit time. The larger the MIDP is, the wider the fault propagation will become. As an indicator of propagation speed and coverage of fault, MIDP is greatly affected by infection and cure rate. If infection rate is high, the fault will quickly spread, and it may cause large-scale cascading failures without proper control. If cure rate is small, the same applies. In particular, the cascading failure can be maintained at the initial state and infected nodes will not grow explosively if infection rate is controlled at a low level and cure rate at a high level. Therefore, the dispatching department can focus on the infection and cure rate to have an effective control over cascading failure.

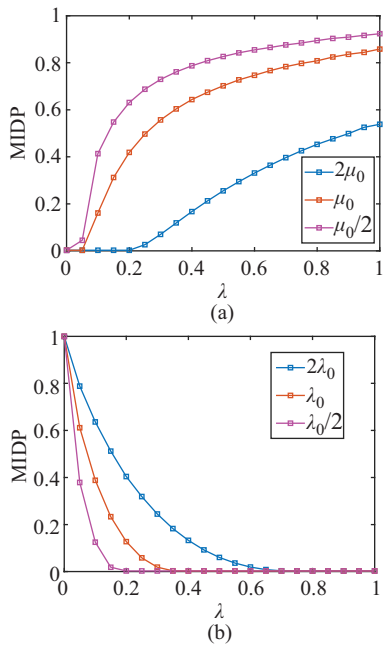

Fig. 12. Comparison of peak density of infected nodes. (a) The infection rate varies from 0 to 1. (b) The cure rate varies from 0 to 1.

2) Comparison of Different Renewable Energy Penetration Rates

In order to compare the impact of renewable energy penetration rate on cascading failure, eleven cases are considered in which wind farms penetration rate increases from 0 to 100% by 10% each case. The fault setting is the same as that in Section V-A, and the other conditions remain unchanged. MIDP and system instability are shown in Table III.

TABLE III
COMPARISON OF THE RESULTS BETWEEN ELEVEN CASES

Number	Penetration rate	MIDP	Whether the system is stable
1	0%	0.8017	Yes
2	10%	0.8324	Yes
3	20%	0.8588	Yes
4	30%	0.9687	Yes
5	40%	0.9722	Yes
6	50%	0.9831	Yes
7	60%	1	No
8	70%	1	No
9	80%	1	No
10	90%	1	No
11	100%	1	No

According to Table III, when the renewable energy penetration rate is between 0% and 60%, MIDP and the risk of cascading failure increase with the renewable energy penetration. When the renewable energy penetration exceeds 60%, the power flow of the system does not converge in the absence of disturbance, and the risk of cascading failures significantly increases. The above phenomena can be explained from the following two aspects.

On the one hand, the increase of renewable energy penetration rate will reduce inertia, thus causing more drastic changes of frequency, power flow and voltage under the same disturbance. Simultaneously, the fault will have a greater impact on the nodes and the infection node density will also increase. On the other hand, when the renewable energy penetration rate increases, the aggregation effect of large-scale energy leads to complex and changeable fault development patterns. The superposition of different fault processes will further aggravate the fault hazards, thus prolonging the time of cascading failures and incurring a larger fault scale and a higher hazard level.

3) Comparison of Different Randomness

In order to consider the randomness of renewable energy, the wind speed is simulated under the scenario that the wind farms penetration rate is about 20%, the wind speed is kept at 11 m/s in the constant wind speed scenario, and the wind speed is set to fluctuate randomly in the range of 8–14 m/s. The relationship between wind power output and wind speed refers to [15]. Other conditions remain unchanged and the comparison of infected nodes density with and without random wind speed is shown in in Fig. 13.

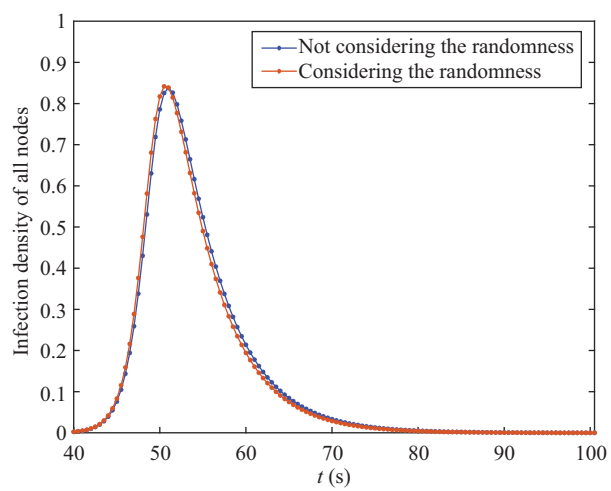

Fig. 13. Density of infected nodes in two cases.

It can be seen from Fig. 13 that with the consideration of the random wind speed the density of infected nodes becomes different, but the impact of a single specific cascading failure process on the result is not obvious. Compared with the load surge disturbance and the fault disturbance, the random fluctuation disturbance of wind speed has less influence on the system, and only a few nodes change their states to reach the threshold.

F. IEEE 118-bus System Testing

In order to further illustrate the effectiveness of the proposed method, IEEE 118-bus system is selected as the object for simulation verification. In this case, the conventional power plants at node 80 and node 89 are replaced by two wind farms of equal capacity respectively, and the total penetration rate of renewable energy is about 11%.

The load active power of nodes 11, 18 and 70 changes to 140 MW, 120 MW and 132 MW at 40 s (2 times the original power). The load of other nodes remains unchanged, and the system load increases by 196 MW. We assume that branches 11–12, 19–20 and 70–71 have transient single-phase ground

fault due to overload at 50 s, and the fault is cleared after 0.5 s. At 55 s, the active power of the three nodes recovers to 70 MW, 60 MW and 66 MW, and the system operation status remains stable.

From 40 s to 50 s, as the 196 MW active load increases, the added load is shared by the generator near the disturbance area, so the active power of branches near the disturbance area is disturbed and the voltage near the disturbance area drops and gradually stabilizes at a lower value than the initial. The node voltage fluctuation causes the disturbance of the reactive power of branches, but the variation of the reactive power of branches is ignorable. When $t = 50$ s, due to the single-phase ground fault on branches 11–12, 19–20 and 70–71, the voltage near the disturbance area slumps suddenly, thus causing the redistribution of the active power in the disturbance area and the change of active power of branches. To maintain the voltage of each node in the system, the generator near the disturbance area increases the excitation current to emit more reactive power, so the reactive power of the branch increases sharply. During 50.5 s to 55 s, due to fault clearance, the active power of each branch gradually stabilizes at a value higher greater than the initial active power. The voltage of the nodes near the disturbance area rises to a stable value, which is lower than the initial, and the change of reactive power of branches is negligible. After 55 s, the active load returns to the original state and the power of each branch returns to the initial value. The voltage of nodes near the disturbance area rises to the initial value and the whole system returns to the initial stable state.

The continuous time calculation of the system is carried out and the computational time is 13.47 s. The comparison of infected nodes density between actual situation and prediction is shown in Fig. 14.

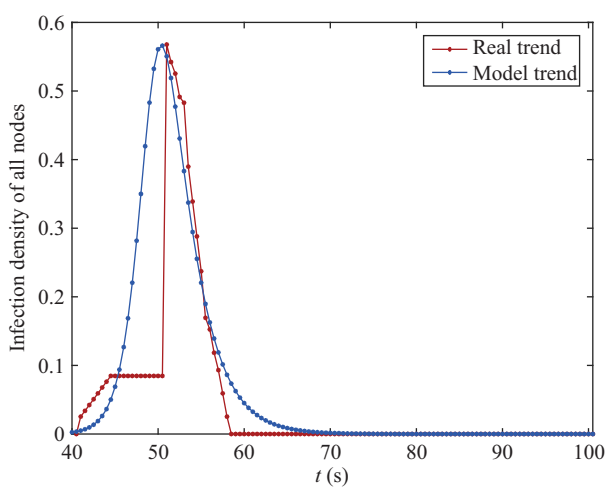

Fig. 14. Infected nodes density of actual and predicted value in IEEE 118-bus system.

Similar to the results in IEEE 39-bus system, the predicted fault propagation is close to the real fault propagation trend, and the arrival time of MIDP is earlier than the real trend. The results again show that the detection method of cascading failure proposed in this paper can achieve an overall advance judgment. Different from the case in IEEE 39-bus system, the fault propagation does not make all nodes of the system infected. There are more generator nodes and load nodes in IEEE 118-bus system; the power supply area is larger; the equivalent grid strength is higher, and the disturbance can be adjusted in the local area.

VI. Conclusion

In this paper, a detection method for development of cascading failure is proposed to solve the failure propagation problem when the proportion of renewable energy keeps getting higher.

1) This method discusses the low inertia of power systems with high penetration of renewable energy and the randomness of renewable energy. In particular, the electrical features, topology features, and timing features are considered in the fault severity index. The proper utilization of the game AHP entropy weight method conduces to online analysis and prediction of fault propagation based on various influencing factors. By comparing with the traditional method and SI epidemic model, the proposed fault propagation dynamic model can reflect the importance of nodes and the impact of faults on each node. It can also provide early warning of failure propagation, which can significantly reduce the risk of cascading failures if key mitigation strategies are implemented in advance. Numerical results on IEEE 39-bus and 118-bus systems demonstrate the effectiveness of the proposed method.

2) This method can be utilized to analyze the severity and propagation mechanism of cascading failure, and it has potential for the control of the spread of cascading failures and the avoidance of the occurrence of large blackouts. With the increase of renewable energy penetration, MIDP also increases under the same fault. MIDP and the infected nodes' density can be used to more comprehensively explore the propagation mode and evolution law of cascading failures under different renewable energy penetration rates. It also has a positive effect on the energy structures safe transformation of the power network.

3) This method can detect the overall situational awareness of cascading failures, quickly locate the fault position and trace its cause in real-time. The key threshold of the node transition state can be obtained by the analogy with the basic contagion number. Based on this, grid operators can obtain the severity of different cascading failure development stages and the effectiveness of different blocking strategies to suppress cascading failures. In this way, the feasibility evaluation of online active security and stability control of power systems with high penetration of renewable energy is realized.

Current studies revolve around the situational awareness of cascading failure in power systems without the intervention of containment strategies. Further research could focus on the identification of the key blocking points in cascading failure and the analysis of the inhibitory effects of different control strategies and their coordination mechanisms.

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Jian Xu received the B.S. and Ph.D. degrees in Electrical Engineering from Wuhan University, Wuhan, China, in 2002 and 2007, respectively. He is currently a Professor with the School of Electrical Engineering and Automation, Wuhan University. His research interests include power system operation, voltage stability, and wind power control and integration.

Zhonghao He received the B.S degree in Electrical Engineering and automation from North China Electric Power University (NCEPU), Beijing, China, in 2017. He is currently pursuing the Ph.D. degree in Civil and Hydraulic Engineering at the School of Electrical Engineering and Automation, Wuhan University. His research interests include power system reliability and cascading failures.

Siyang Liao received the B.S. and Ph.D. degrees in Electrical Engineering from Wuhan University, Wuhan, China, in 2011 and 2016, respectively. He is currently an Associated Professor with the School of Electrical Engineering and Automation, Wuhan University. His research interests include wind power integration and power system simulation technology.

Yuanzhang Sun received the Ph.D. degree in Electrical Engineering from Tsinghua University, Beijing, China, in 1988. He is currently a Professor with the School of Electrical Engineering and Automation, Wuhan University, and a Chair Professor with the Department of the State Key Laboratory of Power System Control and Simulation, Tsinghua University. His research interests include power system dynamics and control, voltage stability and control, and reliability.

Liangzhong Yao received the M.S. and Ph.D. degrees in Electrical Engineering from Tsinghua University, Beijing, China, in 1989 and 1993, respectively. He is currently a Professor with the School of Electrical Engineering and Automation, Wuhan University. His research interests include the integration technology of large-scale renewable energy, operation and control of AC-DC and DC grids.

Deping Ke received the B.S. degree in Electrical Engineering from Huazhong University of Science and Technology, Wuhan, China, in 2005, and the Ph.D. degree in Electrical Engineering from Hong Kong Polytechnic University, Hong Kong, China, in 2012. He is currently an Associated Professor with the School of Electrical Engineering and Automation, Wuhan University. His research interests include power system dynamic and control, and economical operation of power systems.

Jun Yang received the Ph.D. degree in Electrical Engineering from Huazhong University of Science and Technology, Wuhan, China, in 2006. He is currently a Professor with the School of Electrical Engineering and Automation, Wuhan University. His research interests include power system security, electrical vehicles, and smart gird.